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ECE 4644 / 5610 Satellite Communications

Homework Assignment #6 - SOLUTIONS

Problem #1 Ku band VSAT satellite system design

In this question you are asked to analyze the performance of a military satellite communications system that operates in Ku band using a satellite in GEO orbit to connect small, mobile ground stations that are distributed over an area located in the equatorial zone that is the size of a small country. One station is designated the hub while the others operate in receive-only mode. All antennas are the one size. The stations can take turns acting as the hub.

Link and satellite characteristics	
Uplink frequency	15.0 GHz
Downlink frequency	10.0 GHz
Atmospheric clear air loss on uplink	0.6 dB
Atmospheric clear air loss on downlink	0.4 dB
Edge of antenna beam loss on uplink	3.0 dB
Edge of antenna beam loss on downlink	3.0 dB
Miscellaneous losses on uplink	2.0 dB
Miscellaneous losses on downlink	2.0 dB
Satellite system noise temperature	500 K
Satellite transponder bandwidth	48 MHz
Transponder saturated output power	100 W
Transponder back-off	2 dB
Physical temperature of the air, T_p	270^o K
Antenna parameters	
Antenna diameter	0.7 m
Aperture efficiency	60 %
Rx parameters	
Receiver system noise temperature (clear air)	125 K
Receiver noise bandwidth	3.0 MHz
Performance Specification	
Minimum overall C/N	15.0 dB

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Assume the uplink and downlinks have lengths of 36,000 km.

a. Provide a clear, reasonably large, well-labelled sketch of the problem. Show all useful information.

b. Find the gains of the antennas and estimate their beamwidths.

c. Estimate the size of the smaller of the two spots on the surface of the Earth that are covered by the satellite antenna beams when they are directed at the subsatellite point.

d. Calculate the path losses for the uplink and the downlink, in dB.

e. With reference to the uplink, determine the transmit power at the hub station that is required if the received power at the input to the satellite transponder must be at least P_{rx} = -115 dBW. Set out your calculation in the form of a vertical link budget, in dB units. Include losses.

f. Calculate the noise power referred to the input of the transponder, in dBW, in a noise bandwidth of 3 MHz and use this to determine the uplink C/N.

g. Explain why the receiver noise bandwidth of 3 MHz is the correct one to apply in the previous calculation and not the satellite transponder bandwidth of 48 MHz.

h. The receiver is equipped with a high gain LNA. Explain how you can deduce from the given information that the LNA noise temperature is about 100 K. (Hint: solve for the sky noise temperature by using the noise model with G_l for a lossy medium and assuming the antenna temperature is equal to the sky temperature.)

i. Using the information given in the Table, including losses, calculate the received power at the input of a receiving station using a vertical link budget, the noise power in the receiver, and the downlink C/N, all in dB units.

j. Determine the overall C/N for the link and find the clear-air link margin. (Remember that to apply the reciprocal formula the C/N values must be converted to linear units.)

Problem #2

We will continue the analysis of the Ka band VSAT satellite system that was begun with the previous problem. Adopt these values for the clear air situation (may be different from your results):

$(C/N)_{up}$ = 21.8 dB

$(C/N)_{dn}$ = 21.8 dB

$(C/N)_o$ = 18.8 dB

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a) Heavy rain affects the uplink to the satellite causing 3.5 dB of rain attenuation. The downlink is still operating in clear air conditions. What are the new values of $(C/N)_{up}$ and $(C/N)_{dn}$? Find the corresponding overall C/N in the VSAT receiver and compare with the minimum overall C/N = 15 dB (still). Assume linear transponder operation.

b) Now the uplink is clear of rain but the downlink is suffering 3.5 dB of rain attenuation. Find the new value of $(C/N)_{dn}$ and the corresponding overall C/N in the VSAT receiver. Assume that the antenna noise temperature is equal to the sky noise temperature. (Hint: to determine the increase in noise power on the downlink, apply the noise model with G_l, for the combination of clear air attenuation and rain attenuation to obtain the sky noise temperature in the rain. Then determine the system noise temperature in the rain and compare with its clear air value.)

$\Sigma = 25$

c) Finally, determine the amount of rain attenuation on the downlink (to the nearest 0.1 dB) that, in effect, uses up all of the clear air link margin and establishes the limit of link operation for downlink rainfall conditions.

PROBLEM 1

In this question you are asked to analyze the performance of a military satellite communications system that operates in Ku band using a satellite in GEO orbit to connect small, mobile ground stations that are distributed over an area located in the equatorial zone that is the size of a small country. One station is designated the hub while the others operate in receive-only mode. All antennas are the one size. The stations can take turns acting as the hub.

Link and satellite characteristics

Uplink frequency	15.0 GHz
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Antenna diameter	0.7 m
Aperture efficiency	60%

Rx parameters

Receiver system noise temperature (clear air)	125 K
Receiver noise bandwidth	3.0 MHz

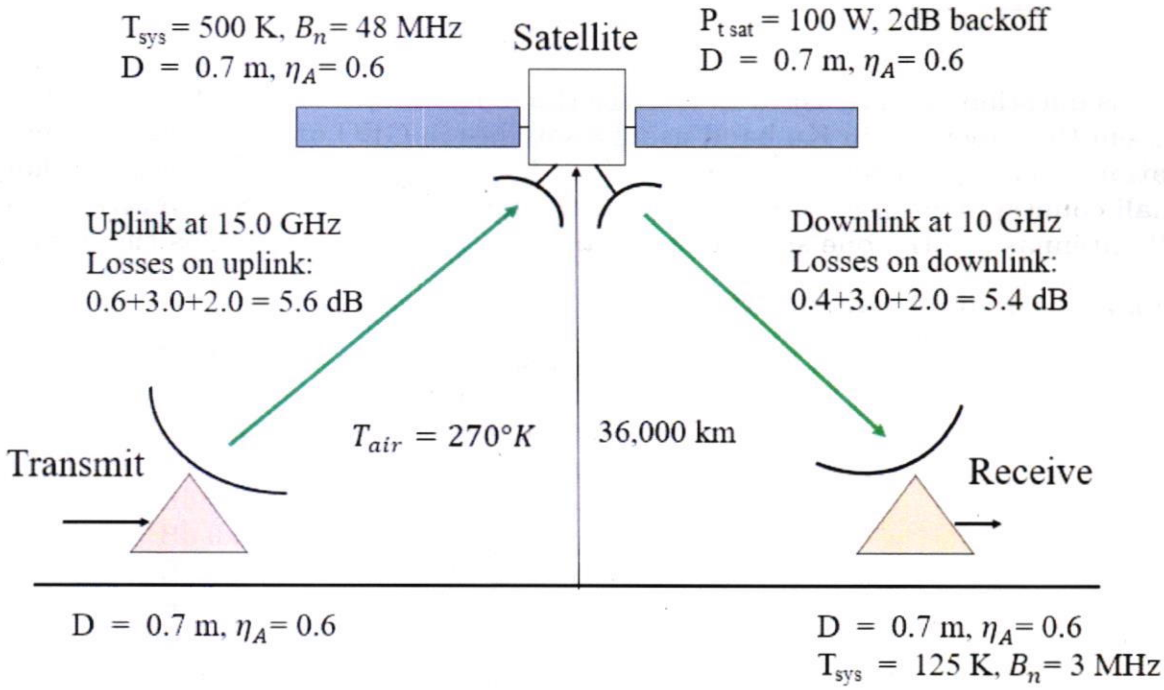
Performance Specification

Minimum overall C/N	15.0 dB

Assure the uplink and downlinks have lengths of 36,000 km.

a. Provide a clear, reasonably large, well-labeled sketch of the problem. Show all useful information.

A sketch of the problem, including all parameters given above, is shown below.



$(C/N)_o \geq 15.0$ dB

b. Find the gains of the antennas and estimate their beamwidths.

The equation used to calculate antenna gain is given below, as are the wavelengths for the uplink and downlink frequencies:

$$G = \eta_A \times \left(\frac{\pi D}{\lambda}\right)^2$$

$$f_{up} \times \lambda = 30 \longrightarrow \lambda = \frac{30}{15} = 2 \text{ cm} = 0.02 \text{ m}$$

$$f_{down} \times \lambda = 30 \longrightarrow \lambda = \frac{30}{10} = 3 \text{ cm} = 0.03 \text{ m}$$

For the antennas operating at the uplink frequency, the gain is calculated and the beamwidth is estimated as shown below:

$$G = 0.6 \times \left(\frac{0.7\pi}{0.02}\right)^2 = 7254.16 = 10\log_{10}(7254.16) = 38.6 \text{ dB}$$

$$\theta_{3dB} \approx \frac{75\lambda}{D} = \frac{75 \times 0.02}{0.7} = 2.14 \text{ deg}$$

For the antennas operating at the downlink frequency, the gain is calculated and the beamwidth is estimated as shown below:

$$G = 0.6 \times \left(\frac{0.7\pi}{0.03}\right)^2 = 3224.07 = 10\log_{10}(3224.07) = 35.1 \text{ dB}$$

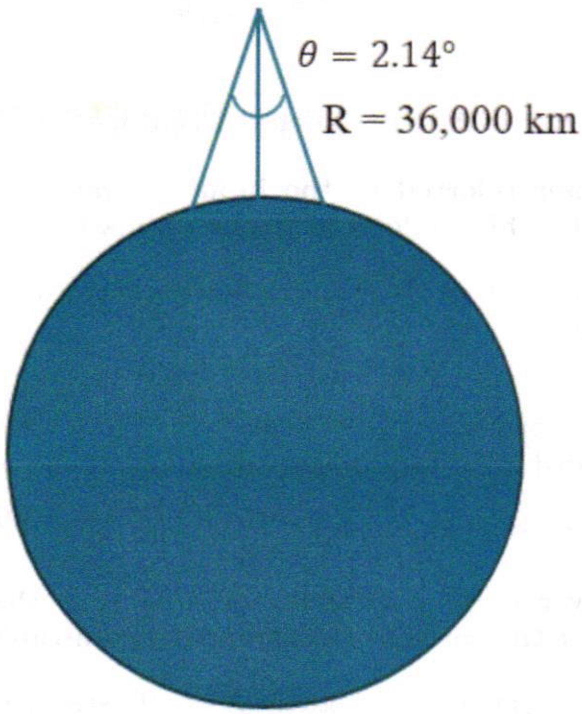
$$\theta_{3dB} \approx \frac{75\lambda}{D} = \frac{75 \times 0.03}{0.7} = 3.21 \text{ deg}$$

c. Estimate the size of the smaller of the two spots on the surface of the Earth that are covered by the satellite antenna beams when they are directed at the subsatellite point.

The smaller beamwidth is 2.14° = 0.03735 rad. Using the visualization below, we can see that the radius of the spot size can be found using trigonometry:

$$r = R \times \tan(\theta/2) = 36000 \times \tan(0.018675) = 672.378 \text{ km}$$

Therefore, the spot size will have a diameter of **1345 km**. Using the same process, the larger beamwidth would produce a spot size of 2017 km.



d. Calculate the path losses for the uplink and downlink, in dB.

The path loss can be calculated using the following equation:

$$L_P = \left(\frac{4\pi R}{\lambda}\right)^2 \longrightarrow L_P = 10\log_{10}\left(\frac{4\pi R}{\lambda}\right)^2 \text{ dB}$$

For the uplink, the distance is 36,000 km and the wavelength is 0.02 m:

$$L_P = 10\log_{10}\left(\frac{4\pi(36 \times 10^6 \text{ m})}{0.02 \text{ m}}\right)^2 \text{ dB} = 207.1 \text{ dB}$$

For the downlink, the distance is 36,000 km and the wavelength is 0.03 m:

$$L_P = 10\log_{10}\left(\frac{4\pi(36 \times 10^6 \text{ m})}{0.03 \text{ m}}\right)^2 \text{ dB} = 203.6 \text{ dB}$$

e. With reference to the uplink, determine the transmit power at the hub station that is required if the received power at the input to the satellite transponder must be at least $P_{rx} = -115$ dBW. Set out your calculation in the form of a vertical link budget, in dB units. Include losses.

The transmit power can be calculated using the link budget equation below:

$$P_r = P_t + G_t + G_r - L_p - L_{misc}$$

The total losses on uplink are $0.6 + 2.0 + 3.0 = 5.6$ dB. The antennas have the same gain and are operating at the uplink frequency for this part of the problem. Using the values previously calculated, the vertical link budget is arranged as shown:

	P_{tx}	P_{tx}
+	G_t	38.6
+	G_r	38.6
-	L_p	207.1
-	L_{tot}	5.6
	P_{rx}	-115

Solving for P_{tx}, the transmit power at the hub station is 20.5 dB, or ≈ 112 W.

f. Calculate the noise power referred to the input of the transponder, in dBW, in a noise bandwidth of 3 MHz and use this to determine the uplink C/N.

The noise power referred to the input of the transponder is calculated using the system noise temperature and the given noise bandwidth:

$$N_{xp} = k + T_{xp} + B_n \text{ dBW}$$

$N_{xp} = -228.6 \text{ dBW/K/Hz} + 10\log_{10}(500 \text{ K}) + 10\log_{10}(3 \times 10^6 \text{ Hz}) = -228.6 + 26.99 + 64.77 = \mathbf{-136.8 \text{ dBW}}$

The uplink C/N can be calculated from the previously determined noise power and the receive power:

$$(C/N)_{up} = P_{rx} - N_{xp} = -115 - (-136.8) = 21.8 \text{ dB}$$

g. Explain why the receiver noise bandwidth of 3 MHz is the correct one to apply in the previous calculation and not the satellite transponder beamwidth of 48 MHz.

The receiver noise bandwidth is essentially the "bottleneck" for the system as it is the smallest noise bandwidth present in the link. While additional noise might be present outside of this bandwidth, it does not need to be included in the overall calculation because it will be filtered out when the signal reaches the receiver.

usually set by the IF filter

h. The receiver is equipped with a high gain LNA. Explain how you can deduce from the given information that the LNA noise temperature is about 100 K. (Hint: solve for the sky noise temperature by using the noise model with G_l for a lossy medium and assuming the antenna temperature is equal to the sky temperature.)

The receiver system noise temperature is the sum of the noise temperatures from the antenna and the LNA:

$$T_{sys} = T_{ant} + T_{LNA}$$

The noise temperature of the antenna is assumed to be equal to the sky temperature. As the physical sky temperature and clear air loss are known, the noise temperature of the sky can be calculated:

$$T_{no} = T_p(1 - G_l) \longrightarrow T_{sky} = 270(1 - 10^{-0.4/10}) = 23.76 \text{ K} = T_{ant}$$

Rearranging to solve for the temperature of the LNA yields the following expression, which shows that the LNA noise temperature is about 100 K:

$$T_{LNA} = T_{sys} - T_{ant} = 125 - 23.76 = 101.24 \text{ K} \approx 100 \text{ K}$$

i. Using the information given in the Table, including losses, calculate the received power at the input of a receiving station using a vertical link budget, the noise power in the receiver, and the downlink C/N, all in dB units.

The total losses on downlink are $0.4 + 2.0 + 3.0 = 5.4$ dB. The transmit power can be converted to dB units from the given saturated output power: $P_t = 100 \text{ W} = 20 \text{ dBW}$.

	P_{tx}	20
-	Backoff	2.0
+	G_t	35.1
+	G_r	35.1
-	L_p	203.6
-	L_{tot}	5.4
	P_{rx}	-120.8 dB

W

The noise power in the receiver is calculated below:

$$N_{xp} = k + T_{xp} + B_n \text{ dBW}$$

$N_{xp} = -228.6 \text{ dBW/K/Hz} + 10\log_{10}(125 \text{ K}) + 10\log_{10}(3 \times 10^6 \text{ Hz}) = -228.6 + 20.97 + 64.77 = -142.9 \text{ dBW}$

The downlink C/N can be calculated from the previously determined noise power and the receive power:

$$(C/N)_{down} = P_{rx} - N_{xp} = -120.8 - (-142.9) = 22.1 \text{ dB}$$

j. Determine the overall C/N for the link and find the clear-air link margin. (Remember that to apply the reciprocal formula the C/N values must be converted to linear units.)

The overall C/N is found using the reciprocal formula:

$$\frac{1}{(C/N)_o} = \frac{1}{(C/N)_{up}} + \frac{1}{(C/N)_{dn}} = \frac{1}{151.356} + \frac{1}{162.181} \longrightarrow (C/N)_o = 78.29 \text{ linear} = 18.9 \text{ dB}$$

The clear-air link margin is $18.9 - 15 = 3.9$ dB.

Problem 2

We will continue the analysis of the Ka band VSAT satellite system that was begun with the previous problem. Adopt these values for the clear air situation (may be different from your results):

$(C/N)_{up} = 21.8$ dB
$(C/N)_{dn} = 21.8$ dB
$(C/N)_o = 18.8$ dB

a. Heavy rain affects the uplink to the satellite causing 3.5 dB of rain attenuation. The downlink is still operating in clear air conditions. What are the new values of $(C/N)_{up}$ and $(C/N)_{dn}$? Find the corresponding overall C/N in the VSAT receiver and compare with the minimum overall $C/N = 15$ dB (still). Assume linear transponder operation.

The problem statement gives the uplink rain attenuation as 3.5 dB. Assuming linear transponder operation as stated, the transmit and receive power will both be affected by the attenuation, therefore affecting the C/N ratios as shown:

$$P_{r,\ rain} = P_{r,\ clear\ air} - A_{up}$$

$$(C/N)_{up,\ rain} = (C/N)_{up} - A_{up} = 21.8 - 3.5 = 18.3 \text{ dB or } 67.61 \text{ (linear)}$$

$$P_{t,\ rain} = P_{t,\ clear\ air} - A_{up}$$

$$(C/N)_{dn,\ rain} = (C/N)_{dn} - A_{up} = 21.8 - 3.5 = 18.3 \text{ dB or } 67.61 \text{ (linear)}$$

$$\frac{1}{(C/N)_o} = \frac{1}{(C/N)_{up}} + \frac{1}{(C/N)_{dn}} = \frac{1}{67.61} + \frac{1}{67.61} \longrightarrow (C/N)_o = 33.81 \text{ linear} = 15.3 \text{ dB}$$

With rain attenuation included, the overall C/N is reduced to barely above the minimum value by 0.3 dB. The uplink rain attenuation of 3.5 dB is close to the uplink margin of 3.8 dB (found by taking the difference between the minimum C/N and the overall clear air C/N).

b. Now the uplink is clear of rain but the downlink is suffering 3.5 dB of rain attenuation. Find the new value of $(C/N)_{dn}$ and the corresponding overall C/N in the VSAT receiver. Assume that the antenna noise temperature is equal to the sky noise temperature. (Hint: to determine the increase in noise power on the downlink, apply the noise model with G_l, for the combination of clear air attenuation and rain attenuation to obtain the sky noise temperature in the rain. Then determine the system noise temperature in the rain and compare with its clear air value.)

The sky noise temperature in the rain is calculated below, including both the atmospheric loss on downlink and the downlink rain attenuation:

$$T_{sky\ rain} = 270(1 - G_l) = 270(1 - 10^{-(0.4+3.5)/10}) = 270(1 - 10^{-(3.9)/10}) = 160 \text{ K} = T_{ant}$$

The noise temperature of the LNA was previously found to be 101 K in Problem 1 Part H, and We know from the given information that the receiver system noise temperature in clear air is 125 K:

$$T_{s\ rain} = T_{sky\ rain} + T_{LNA} = 160 + 101 = 261 \text{ K}$$

The increase in noise power in the receiver is calculated below. This is used with the rain attenuation and clear air C/N values to find the new downlink C/N:

$$\Delta N = 10\log(261/125) = 3.2 \text{ dB}$$

$$(C/N)_{dn} = (C/N)_{dn} \text{ clear air} - A_{dn} - \Delta N = 21.8 - 3.5 - 3.2 = \boxed{15.1 \text{ dB}}$$

The power received at the transponder input is the same as in clear air, so the uplink C/N stays the same. The overall C/N can now be calculated:

$$\frac{1}{(C/N)_o} = \frac{1}{(C/N)_{up}} + \frac{1}{(C/N)_{dn}} = \frac{1}{151.356} + \frac{1}{32.359} \longrightarrow (C/N)_o = 33.81 \text{ linear} = \boxed{14.3 \text{ dB}}$$

This value is less than the minimum overall C/N of 15 dB, meaning that this level of downlink rain attenuation will result in an outage.

c. Finally, determine the amount of rain attenuation on the downlink (to the nearest 0.1 dB) that, in effect, uses up all of the clear air link margin and establishes the limit of link operation for downlink rainfall conditions.

The MATLAB algorithm below was written to iteratively increase the downlink rain attenuation until the overall link margin is no longer 15 dB.

```
clc
clear
close all

T_LNA = 101; % lna noise temp
T_sys_clear = 125; % clear air receiver system noise temp
A_clear = 0.4; % clear air attenuation
CNR_clear = 21.8; % clear air downlink C/N
CNR_up = 21.8; % uplink C/N

A_rain = 1; % initialize downlink rain attenuation

while true
    T_sky_rain = 270*(1-(10^(-(A_clear+A_rain)/10))); % sky noise temp w/ rain
    T_sys_rain = T_sky_rain + T_LNA; % system noise temp w/ rain
    Del_N = 10*log10(T_sys_rain/T_sys_clear); % noise value increase
    CNR_rain = CNR_clear - A_rain - Del_N; % downlink C/N in rain

    CNR_o = pow2db(1/((1/db2pow(CNR_up))+(1/db2pow(CNR_rain)))); % overall C/N
```

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    % terminate loop if link no longer closes
    if CNR_o < 15
        break
    end

    A_rain = A_rain+0.1; % iterate rain attenuation
end
```

For $A_{dn} = 2.8$ dB, $(C/N)_o = 15.09$ dB. For For $A_{dn} = 2.9$ dB, $(C/N)_o = 14.97$ dB. Therefore, the maximum rain attenuation on the downlink is 2.8 dB.